A Feasibility-First Full-Recharge ALNS for the Electric Vehicle Routing Problem with Time Windows

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MOTIVATION AND RESEARCH QUESTION

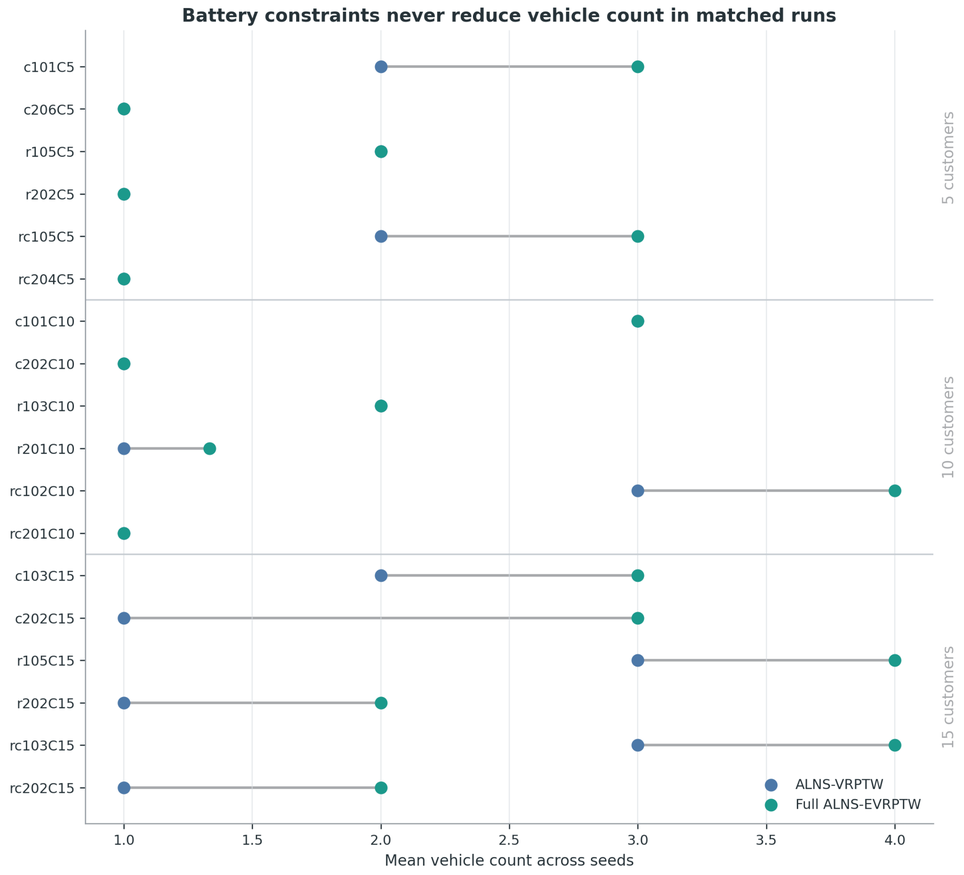
Electric delivery fleets must satisfy customer time windows while keeping every route within battery range. This couples routing, scheduling, and charging decisions.

Can a compact ALNS solver enforce strict EV feasibility and reveal the operational cost of electrification on benchmark instances?

- Compare the same search framework with and without battery constraints.
- Prioritise feasibility first, then minimise fleet size and distance.
- Diagnose which algorithm components contribute useful improvements.

EXPERIMENTAL DESIGN

Benchmark	18 small Schneider EVRPTW instances
Scale	5, 10, and 15 customers
Repetitions	3 random seeds per configuration
Budget	1,000 ALNS iterations per run
Comparison	Five configurations; 270 runs in total
Charging model	Full recharge at duplicated station nodes



Battery constraints increased fleet size in **28/54** matched runs and never reduced it.

MAIN RESULTS

216/216
battery-enabled runs strictly feasible

28/54
matched runs used more EVs

5.65–20.22×
runtime ratio from 5 to 15 customers

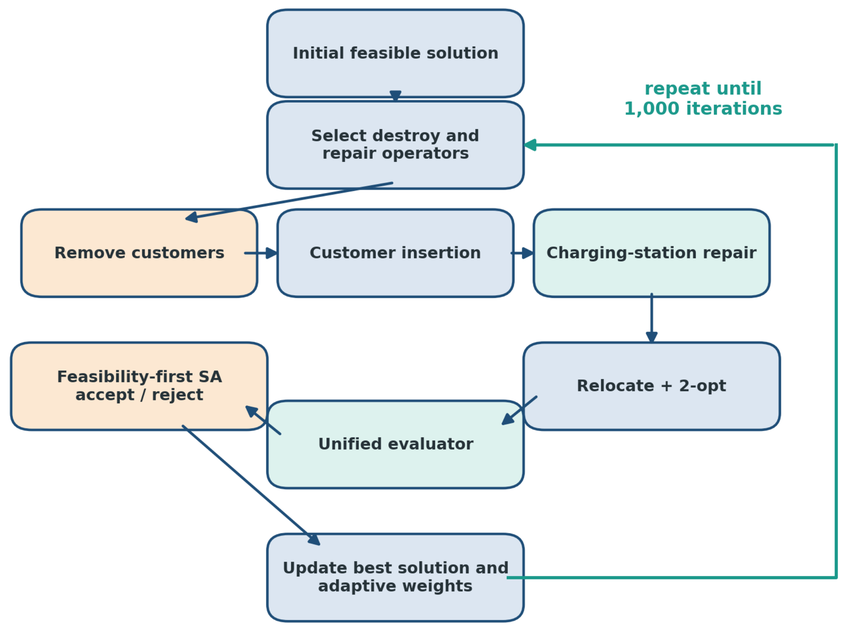
Among equal-fleet comparisons, EV routes were longer in **23/26** cases, with a mean distance penalty of **+12.419**. Charging feasibility therefore raised both route length and computational cost.

FEASIBILITY-FIRST ALNS

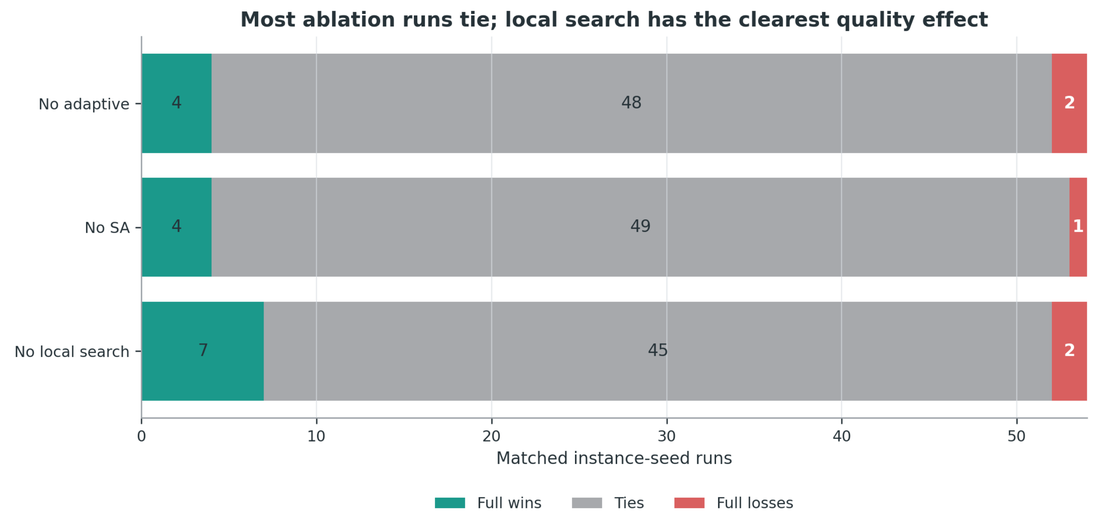
The solver uses a common destroy–repair loop for VRPTW and EVRPTW. Battery-enabled evaluation inserts charging stations greedily, propagates time and energy states, and rejects any route that remains infeasible.

- **Lexicographic objective:** minimise vehicles first, then total distance.
- **Adaptive operators:** random, worst, and Shaw removal with greedy or regret repair.
- **Search control:** simulated annealing acceptance, adaptive weights, and lightweight local search.

Week 6 ALNS-EVRPTW search pipeline



ABLATION: WHAT DID LOCAL SEARCH ADD?



- Full ALNS won 7, tied 45, and lost 2 of 54 paired comparisons.
- Removing local search reduced runtime by **47.3%**.
- **95.6%** of iterations produced equal-objective candidates, indicating limited search diversity at this budget.

CONCLUSION AND LIMITATIONS

The implementation reliably enforced strict full-recharge EV feasibility. On these small instances, electrification often required more vehicles, longer routes, and substantially more runtime.

Scope of the evidence

- Small benchmark subset and three seeds only.
- Full recharge and greedy station insertion; no station-removal move.
- The ablation suggests future work should target charging-specific neighbourhoods rather than more generic local search.

Selected references: Schneider et al. (2014), *Transportation Science*; Pisinger and Ropke (2007), *Computers & Operations Research*.

TAKE-HOME MESSAGE

216/216
strictly feasible EV runs

28/54
comparisons used more EVs

47.3%
less runtime without local search